

Software Requirements Specification for CEWS System: Data Separability Measurement

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Revision History

Date	Version	Notes
2023-01-24	0.0	Creation
2023-02-01	0.1	Revise Instance Models
2023-02-03	1.0	Version 1 Release
2023-04-16	2.0	Revision based on feedback

1 Reference Material

This section records information for easy reference.

1.1 Table of Units

Throughout this document no units are used.

1.2 Table of Symbols

The table that follows summarizes the symbols used in this document along with their units. The choice of symbols was made to be consistent with the statistics literature.

Symbol	Unit	Description
$I(\dots)$	N/A	Indicator Function, returns 1 if its argument is true and 0 otherwise
J_N	N/A	Normalized Multiclass Cut Edge Weight Measurement
n	N/A	Number of observations
m	N/A	Number of features
C	N/A	Class label matrix of size $n \times 1$
D	N/A	Data matrix of size $n \times m$
V	N/A	Graph connection matrix
$d(a, b)$	N/A	Euclidean distance between points a and b

1.3 Abbreviations and Acronyms

Symbol	Description
A	Assumption
CEWS	Cut Edge Weight Statistic
DD	Data Definition
GD	General Definition
GS	Goal Statement
IM	Instance Model
LC	Likely Change
PS	Physical System Description
R	Requirement
RNG	Relative Neighbourhood Graph
SRS	Software Requirements Specification
T	Theoretical Model

1.4 Mathematical Notation

This document uses notation the used in [\[1\]](#) and [\[2\]](#).

2 Introduction

Classification is a common machine learning problem, many techniques are used for solving classification problems. Unfortunately, not all classification problems are easy and it can be difficult to explain why classifiers perform poorly on certain datasets. Data complexity has been defined as a characteristic of the data that relates to the difficulty of the classification problem [3]. To further refine this definition, we consider data complexity to be represented by aspects of the data that introduce difficulty for the classifiers.

One of these aspects is separability, an intrinsic characteristic that describes how the observations from different classes mix together [4]. A classic measurement for separability is Fraction of Points on Class Boundary. Based on the work of Friedman and Rafsky [5] and documented by Basu and Ho [6]. This measurement constructs a minimum spanning tree over the dataset then calculates the number of points with a connection to another class over all points.

The Cut Edge Weight Statistic (CEWS) measurement is a variation of the Fraction of Points on Class Boundary, proposed by Zighed et al. [1]. The algorithm for the normalized measurement begins by constructing a Toussaint’s Relative Neighbourhood Graph [7] over the entire dataset then computing the number of edges that connect different classes (the “cut edges”) over the total number of edges (shown in Figure 1). This measurement is a viable way of measuring separability because it has a high coefficient of determination (R^2) with the mean error rate of several classifiers [1]. CEWS System is a program designed to calculate the CEWS.

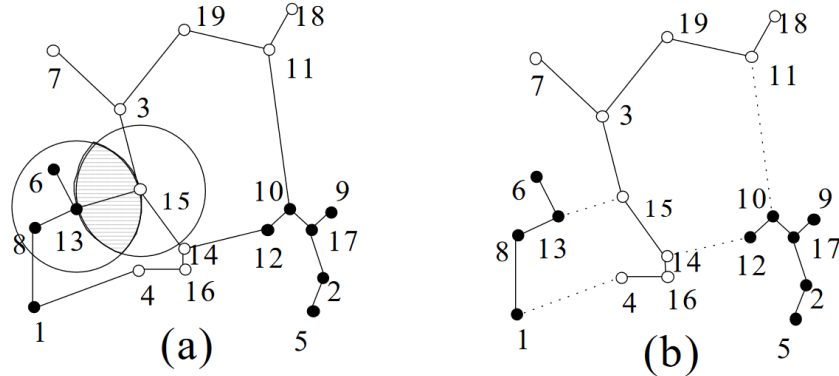


Figure 1: An visualization of the RNG for a binary classification problem represented by black and white points [1]. (a) An example of a 2D RNG graph construction. (b) Identifying cut edges with dotted lines.

The introduction contains the purpose of this document, the scope of requirements, characteristics of the indented reader and the organization of this complete document.

2.1 Purpose of Document

The purpose of this document is to provide information about the design and functionality of CEWS System to the Intended Reader (Section 2.3).

2.2 Scope of Requirements

The domain of this problem is restricted to generating the normalized multiclass CEWS Measurement [1]. Notable exclusions include the calculation of a p -value, variations of the calculation using different weighing parameters and different graph generation algorithms. The scope excludes asymptotic cost limitations and efficiency requirements.

2.3 Characteristics of Intended Reader

The reader should have a good understanding of classification (at a graduate level or equivalent experience). The reader should have a base understanding of data analysis, machine learning and statistics such as from standard undergraduate courses. The reader should be familiar with basic mathematical number types and their common symbols.

2.4 Organization of Document

This document includes a general system description, a specific system description, a requirements section, a section on likely changes, a section on unlikely changes and a section on traceability.

3 General System Description

This section provides general information about the system. It identifies the interfaces between the system and its environment, describes the user characteristics and lists the system constraints.

3.1 System Context

The CEWS System will exist as a standalone product or as a module. The system context is illustrated in Figure 2.

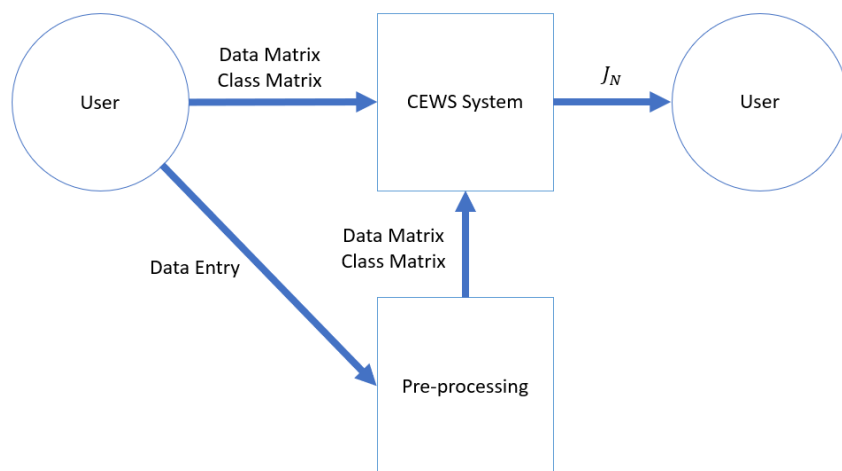


Figure 2: System Context

- User Responsibilities:
 - Provide the input data matrices. (Section [4.2.6](#))
 - Correctly format the input data matrices as per Section [4.2.6](#). (For example, it is the responsibility of the user to make sure the points and labels are correctly associated as this can not be checked by the system.)
 - Manage the computer’s resources including supplying the system with enough resources to support the size of their dataset.
 - Understand the problem.
 - Perform any data pre-processing required for the relevant classification problem such as transformations, feature extraction, etc.
 - Understand the CEWS algorithm.
 - Understand the limitations of numerical computation.
- CEWS System Responsibilities:
 - Read input data.
 - Detect data type mismatch or if input data violates any constraints. (Section [4.2.6](#))
 - Perform the analysis.
 - Perform basic error checking on the output.
 - Return the output value.

3.2 User Characteristics

The end user of CEWS System should have a good understanding of classification and machine learning, preferably with an undergraduate degree in software or mathematics.

3.3 System Constraints

At this time, there are no known constraints for this system.

4 Specific System Description

This section first presents the problem description, which gives a high-level view of the problem to be solved. This is followed by the solution characteristics specification, which presents the assumptions, theories, definitions and finally the instance models.

4.1 Problem Description

CEWS System is intended to solve the calculation of the normalized multiclass CEWS measurement for a dataset.

4.1.1 Terminology and Definitions

This subsection provides a list of terms that are used in the subsequent sections and their meaning, with the purpose of reducing ambiguity and making it easier to correctly understand the requirements:

- Observation: An identifiable element in a data set that has features and a class label associated with it.
- Dataset: A set of observations from a population including information about those observations.
- Feature: An individual measurable property of an observation.
- Class label: Represents the class an observation belongs to.
- Classification: The problem of identifying which of a set of categories an observation belongs to.
- Data complexity: Aspects of the data that classifiers have difficulty with [3].
- Separability: A type of data complexity that measures how the observations from different classes mix together [4].
- Boolean variable: A variable that has one of two possible values.
- Data Matrix: a matrix containing the features for each observation.
- Class Matrix: a matrix containing class labels.

4.1.2 Physical System Description

There is no physical system.

4.1.3 Goal Statements

Given the dataset, the goal statement is:

GS1: Calculate the normalized multiclass CEWS measurement.

4.2 Solution Characteristics Specification

The instance models that govern CEWS System are presented in Section 4.2.5. The information to understand the meaning of the instance models and their derivation is also presented, so that the instance models can be verified.

4.2.1 Assumptions

This section simplifies the original problem and helps in developing the theoretical model by filling in the missing information for the physical system. The numbers given in the square brackets refer to the theoretical model [T], general definition [GD], data definition [DD], instance model [IM], or likely change [LC], in which the respective assumption is used.

- A1: All rational numbers can be represented in a base-2 computer system. Operations performed on rational numbers in a computer system is exact. Any computation rounding error is negligible and does not need to be taken into account.
- A2: All observations from one class share the same class label. (For example, no sub-classes exist.)
- A3: Weights are not used for the edge calculation.
- A4: A relative neighbourhood graph is used as the base graph.
- A5: The statistic p -value is not needed.

4.2.2 Theoretical Models

No theoretical models are used.

4.2.3 General Definitions

This section collects the equations that will be used in building the instance models.

Number	GD1
Label	Toussaint's Relative Neighbourhood Graph
SI Units	N/A
Equation	Connection matrix: $V = (v_{ij}), i = 1, 2, \dots, n; j = 1, 2, \dots, n;$ where $v_{ij} = I([d(p_i, p_j) \leq \max(d(p_i, c), d(p_j, c)) \forall c, c \in P, c \neq p_i, c \neq p_j])$
Description	RNG is an undirected graph constructed by connecting an edge between two points when a third point does not exist that is closer to both. P is a set of n points in real space with dimension(s) m with elements $p_i: i = 1, \dots, n$. V is the connection matrix that represents the RNG where $v_{ij} = 1$ if there is an edge and $v_{ij} = 0$ if not. $d(a, b)$ is the Euclidean distance between points a and b given by DD1. Based on assumptions 1.
Source	[1]
Ref. By	IM1

4.2.4 Data Definitions

This section collects and defines all the data needed to build the instance models. The dimension of each quantity is also given.

Number	DD1
Label	Euclidean distance
Symbol	$d(a, b)$
SI Units	N/A
Equation	$d(a, b) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$
Description	$d(a, b)$ is the Euclidean distance between points a and b in Euclidean space with n dimensions. a_i and b_i are vectors from the origin. Based on assumption 1.
Sources	N/A
Ref. By	GD1

4.2.5 Instance Models

This section transforms the problem defined in Section 4.1 into one which is expressed in mathematical terms. It uses concrete symbols defined in Section 4.2.4 to replace the abstract symbols in the models identified in Sections 4.2.2 and 4.2.3. The goal GS1 is solved by IM1, which is the implementation of the cut edge weight statistic measurement.

Number	IM1
Label	Multiclass Normalized Cut Edge Weight Measurement
Input	The input data matrix, $D_{n \times m}$, and the class label matrix, $C_{n \times 1}$, based on 4.2.6.
Output	J_N , $0 \leq J_N \leq 1$, such that $J_N = \frac{J}{I+J}$ where $J = \sum_{r=1}^{k-1} \sum_{s=r+1}^k J_{r,s}; 0 \leq J \leq n^2$ where $J_{r,s} = \sum_{i=1}^n \sum_{j=1}^n V_{ij} \times I(c_i = r \ \& \ c_j = s)$ $I = \sum_{r=1}^k I_r, 0 \leq I \leq n^2$ where $I_r = \sum_{i=1}^{n-1} \sum_{j=i+1}^n V_{ij} \times I(c_i = r \ \& \ c_j = r)$
Description	Based on assumptions 1, 2, 3, 4 and 5. The above equation gives the formula for calculating the the multiclass normalized cut edge weight measurement. k is the number of classes. V is the RNG connection matrix for matrix D as P given by GD1. v_{ij} is element i, j of V. c_i is element i of C.
Sources	[1] [8]
Ref. By	None

4.2.6 Input Data Constraints

Table 1 shows the data constraints on the input variables. The column for physical constraints gives the physical or theoretical limitations on the range of values that can be taken by the variable. The column for software constraints restricts the range of inputs to reasonable values. The software constraints will be helpful in the design stage for picking suitable algorithms. The constraints are conservative, to give the user of the model the flexibility to experiment with unusual situations. The column of typical values is intended

to provide a feel for a common scenario. The uncertainty column provides an estimate of the confidence with which the physical quantities can be measured.

The user provides a dataset in the format of matrices D and C . n is the number of observations. m is the number of features. The input data matrix, D , is of size $n \times m$ and must contain rational numbers. D is formatted as:

$$D = \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1m} \\ d_{21} & d_{22} & \cdots & d_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & \cdots & d_{nm} \end{bmatrix}$$

So row i , d_i , contains observation i . The input class matrix, C , is of size $n \times 1$ and must contain natural numbers. C is formatted as:

$$C = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Where c_i is the label for observation i of matrix D . It is the user's responsibility to set class label c_i to the class of observation d_i . (Section 3.1) The size of n and m are limited based on the system hardware but not by CEWS itself. It is the user's responsibility to insure that their system has the relevant requirements to support their dataset size.

Table 1: Input Variables

Var	Constraints	Software Constraints	Typical Value	Uncertainty
d_{ij}	$d_{ij} \in \mathbb{R}$	$d_{ij} \in \mathbb{Q}, F_{\min} \leq d_{ij} \leq F_{\max}$	N/A	N/A
c_i	$c_i \in \mathbb{N}$	$W_{\min} \leq c_i \leq W_{\max}$	N/A	N/A
n	$n \in \mathbb{N}, n > 0$	N/A	N/A	N/A
m	$m \in \mathbb{N}, m > 0$	N/A	N/A	N/A

Table 2: Specification Parameter Values

Var	Value
$F_{\min}$	$-(2 - 2^{-23})2^{127}$
$F_{\max}$	$(2 - 2^{-23})2^{127}$
$W_{\min}$	0
$W_{\max}$	$2^8 - 1$

4.2.7 Properties of a Correct Solution

A correct solution must exhibit the following behaviour: Table 3.

Table 3: Output Variables

Var	Physical Constraints
J_N	$0 \leq J_N \leq 1$

5 Requirements

This section provides the functional requirements, the business tasks that the software is expected to complete, and the nonfunctional requirements, the qualities that the software is expected to exhibit.

5.1 Functional Requirements

R1: CEWS System will read the input data.

- Any constraint in 4.2.6 is violated, the CEWS System will generate an appropriate error message.

R2: After reading the input data, CEWS System will perform the calculation in IM1 and output the result.

- In case an error occurs, an error will be raised by the program.
- In case no error occurs, J_N will be the output.

R3: The CEWS System will generate the exact same output for the exact same dataset.

- Including if the order of the dataset changes, as long as the same labels are associated with the same points.

5.2 Nonfunctional Requirements

- NFR1: **Accuracy** The accuracy of the computed solutions should meet the level needed for data analysis. The absolute error should be less than `ACCURACY_ERR_ABS`. (See Table 7) The absolute error is used because the error should not vary based on the size of the result (the scale of the values is limited).
- NFR2: **Portability** The software should run any machine running the Windows 11 operating system.

6 Likely Changes

- LC1: The addition of user options such as option to weigh the edge calculations: IM1. Related to: A3.
- LC2: The addition of user options such as option to change the algorithm for graph construction: IM1. Related to: A4.
- LC3: Changes to the output format. Related to: A5.

7 Unlikely Changes

There are no unlikely changes anticipated.

8 Traceability Matrices and Graphs

The purpose of the traceability matrices is to provide easy references on what has to be additionally modified if a certain component is changed. Every time a component is changed, the items in the column of that component that are marked with an “X” may have to be modified as well. Table 5 shows the dependencies of theoretical models, general definitions, data definitions, and instance models with each other. Table 6 shows the dependencies of instance models, requirements, and data constraints on each other. Table 4 shows the dependencies of theoretical models, general definitions, data definitions, instance models, and likely changes on the assumptions.

	GD1	DD1	IM1
GD1		X	X
DD1	X		
IM1	X		

Table 5: Traceability Matrix Showing the Connections Between Items of Different Sections

	A1	A2	A3	A4	A5
GD1	X		X		
DD1	X				
IM1	X	X	X	X	X
LC1			X		
LC2				X	
LC3					X

Table 4: Traceability Matrix Showing the Connections Between Assumptions and Other Items

	IM1
IM1	
R1	X
R2	X
R3	X
NFR1	X
NFR2	X

Table 6: Traceability Matrix Showing the Connections Between Requirements and Instance Models

9 Development Plan

Not applicable.

10 Values of Auxiliary Constants

The definition of the test cases will call for SYMBOLIC_CONSTANTS. Their values are defined in this section for easy maintenance.

Table 7: Table of SYMBOLIC_CONSTANTS

Var	Value
ACCURACY_ERR_ABS	0.1

References

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